



Dynamic ensemble fault diagnosis framework with adaptive hierarchical sampling strategy for industrial imbalanced and overlapping data

Haoyan Dong^{a,b}, Chuang Peng^{a,b}, Lei Chen^{a,b}, Kuangrong Hao^{a,b}

^a Engineering Research Center of Digitalized Textile & Fashion Technology, Ministry of Education, Donghua University, Shanghai, 201620, China

^b College of Information Science and Technology, Donghua University, Shanghai, 201620, China

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ABSTRACT

The coexistence of class imbalance and class overlap significantly challenges fault diagnosis in modern industrial processes. Class imbalance, characterized by the scarcity of fault data, and class overlap, arising from similarities between normal and fault data as well as correlations among fault types, are intertwined issues that jointly degrade fault diagnosis performance. To address these coupled issues, this paper proposes a dynamic ensemble fault diagnosis framework with adaptive hierarchical sampling strategy (DEAHS). The framework employs a boosting ensemble structure, effectively mitigating class imbalance through dynamic majority class undersampling and reducing class overlap by focusing on minority classes in high-overlap regions. In the outer layer, a Markov decision process guides the adaptive undersampling of majority class, achieving relatively balanced subsets. In the inner layer, a membership entropy-based method identifies overlap regions, and a weighted oversampling strategy improves minority classes' representation in these regions. The proposed framework is validated on the Tennessee Eastman process and a real-world polyester esterification process, where its performance is evaluated using four metrics commonly employed for imbalanced datasets. The results demonstrate that the proposed method achieves superior performance across a majority of metrics, highlighting its effectiveness in handling imbalanced and overlapping industrial fault data.

1. Introduction

Fault diagnosis can identify fault conditions in industrial processes, which is crucial for ensuring the reliability of complex industrial equipment and reducing the economic losses caused by system faults [1,2]. In the past few decades, the increasing complexity of systems has posed challenges to traditional fault diagnosis techniques based on physical models [3], as they typically require accurate mechanistic models and extensive expert knowledge. With the rapid development of advanced sensors, distributed computer systems, and Internet of Things (IoT) technologies, a large amount of monitoring data from industrial processes can be recorded and stored, providing a robust data foundation for data-driven fault diagnosis methods. Therefore, data-driven methods [4–6] have been widely applied in intelligent fault diagnosis of industrial processes.

Many existing data-driven fault diagnosis methods often assume that the dataset follows a normal distribution. However, in real industrial processes, systems operate under stable conditions most of the time, with faults occurring very rarely. This leads to an imbalanced class distribution in the data collected from the distributed control

system (DCS). The data follows a long-tailed distribution, where the majority class consists of data from normal conditions, and multiple minority classes represent data from various fault conditions. Meanwhile, the transition from normal conditions to fault conditions is often not abrupt. In the early stages of a fault, there is a similarity between the fault data and the normal data. Additionally, the complexity of modern industrial systems results in correlations between different faults, with data from various fault types often exhibiting similar characteristics. Therefore, data-driven fault diagnosis in industrial processes faces not only the challenge of class imbalance but also the challenge of class overlap.

Existing studies [7,8] indicate that when class overlap and class imbalance are combined, the performance of classifiers significantly deteriorates. On the one hand, the scarcity of minority-class samples, as a manifestation of class imbalance, makes it difficult to adequately represent their underlying distributions, allowing the decision boundary to be dominated by the majority class and weakening the model's ability to recognize the minority classes. On the other hand, class overlap further intensifies the uncertainty in distinguishing between

* Corresponding authors.

E-mail addresses: donghaoyan@mail.dhu.edu.cn (H. Dong), chuangpeng@mail.dhu.edu.cn (C. Peng), leichen@dhu.edu.cn (L. Chen), krhao@dhu.edu.cn (K. Hao).

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different classes. In industrial processes, the relatively scarce fault samples demand more attention compared to the abundant normal samples. Notably, the early-stage fault samples, which are of high value for system maintenance, are often interwoven and overlapped with numerous normal samples. This coupling of imbalance and class overlap renders the already limited information from minority classes even harder to extract and identify effectively. Under such circumstances, the model becomes more susceptible to the influence of the majority class during training, struggling to accurately capture the intrinsic features of the minority classes and exhibiting high uncertainty when dealing with entangled, ill-defined categories. These challenges greatly hinder the timely and accurate detection of potential faults in industrial processes. Therefore, investigating the coupling of imbalance and class overlap is of significant practical importance in the field of industrial process fault diagnosis.

Class imbalance has garnered widespread attention from researchers [9] in fault diagnosis. These methods can be broadly categorized into three types: data-level methods, model construction methods, and training optimization methods [10]. In data-level methods, researchers alter the number of samples in each class by designing different sampling strategies [11–13] or by using generative models [14,15] to create new samples, thereby constructing a balanced dataset. In model construction methods, researchers often reduce the model's sensitivity to class imbalance by modifying classification algorithms or by incorporating concepts from ensemble learning [16] and transfer learning [17,18]. Additionally, in training optimization methods, researchers mitigate the dominance of the majority class during the training process by developing new training strategies, such as multi-stage training [19,20] or cost-sensitive learning [21].

Although the aforementioned methods effectively address the challenges posed by class imbalance in fault diagnosis, they seldom focus on the issue of class overlap present in imbalanced fault datasets. It is widely recognized that class imbalance is not the only factor impeding classification. The true decline in classification performance occurs when class imbalance combines with other factors, with class overlap being the most detrimental.

For the coupling of imbalance and class overlap, simple and efficient data-level methods are the most popular, compared to model construction methods that rely on specific classifiers or cost-sensitive approaches tailored to particular datasets or problems. The classic oversampling method is based on the synthetic minority oversampling technique (SMOTE) [22]. Subsequently, to overcome its tendency to generate noise, uninformative, boundary, and overlapping samples, researchers have proposed several variants of this method. For example, some variants combine SMOTE with clustering methods, such as OEGFCM-SMOTE [23] and DeepSMOTE [24]. Other approaches, like FW-SMOTE [25] and NaNSMOTE [26], focus on assigning weights to samples based on neighboring samples. The classic undersampling methods are based on nearest neighbor search, such as Tomek link [27] and its variants [28], which are also well-suited for scenarios involving class overlap. Dai et al. [29] proposed a multi-granularity relabeled under-sampling algorithm (MGRU) based on Tomek-Link, which leverages local sample information and detects overlapping samples. Some researchers have also taken a global approach by proposing an undersampling method for class overlap based on Schur matrix decomposition, known as SDCU [30]. These existing studies demonstrate that oversampling can effectively address class imbalance, while undersampling is more suitable for resolving class overlap.

Therefore, hybrid sampling is a natural approach for addressing the combination of class imbalance and overlap. For example, Mirzaei et al. [31] proposed a clustering and density-based hybrid (CDBH) sampling method, while Zhu et al. [32] introduced an evolutionary hybrid sampling technique. These methods commonly involve undersampling to remove irrelevant majority class samples and applying strategies to prevent oversampling from creating noisy or overlapping samples, leading to a clearer decision boundary for the classifier. Recent

studies have highlighted the combination of data-level approaches with model-building methods or training strategy optimizations, effectively leveraging the strengths of different techniques and becoming a research focus [33,34]. Ensemble learning, in particular, has attracted considerable interest from researchers. The core idea is to use different sampling methods to obtain various subsets for training multiple base classifiers, leading to a more reliable classification model. Ensemble methods can be divided into bagging and boosting. Bagging methods involve parallel training, such as UnderBagging [35]. Boosting methods are trained sequentially, including SPE [36] and MESA [37].

Unlike the straightforward imbalance ratio (IR) [7] used to measure class imbalance, class overlap is more complex and requires diverse methods for definition and measurement. Addressing class overlap involves three steps: (1) decomposing the data space, (2) identifying overlapping regions, and (3) applying tailored methods to handle these regions based on the task. Establishing specific methods to address overlapping regions after identification undoubtedly helps enhance the separability between classes. However, in industrial process fault diagnosis, most existing methods address only class imbalance, neglecting the combined effects of overlap and imbalance. While some claim their algorithms are robust to class overlap and noise, few specifically address overlapping regions.

To address these limitations, this paper proposes a dynamic ensemble fault diagnosis framework with adaptive hierarchical sampling strategy (DEAHS). The framework is designed to address the coupling of class imbalance and overlap in industrial fault datasets, which are common challenges in real-world applications. In such scenarios, normal operating conditions dominate as the majority class, while minority fault classes not only face severe imbalance but also exhibit significant overlap at class boundaries, making accurate and robust fault diagnosis particularly challenging. The framework builds an ensemble classifier through boosting, utilizing adaptive hierarchical hybrid sampling to generate balanced and well-separated subsets. The outer layer models the majority class sampling process as a Markov Decision Process (MDP), enabling a trained sampler to dynamically adjust sampling weights based on classification errors, thereby reducing imbalance while preserving diversity. The inner layer tackles class overlap by dividing data into safe and overlapping regions using membership entropy and applying weighted oversampling to minority samples in overlapping regions. Sampling weights are determined by the contributions of these samples to the between-class scatter matrix, enhancing class separability while further mitigating residual imbalance. By integrating these two adaptive layers, the DEAHS framework provides a unified solution to class imbalance and overlap, improving fault diagnosis performance in challenging industrial scenarios.

The main contributions can be summarized as follows:

- (1) The DEAHS framework is proposed to address the coupling of class imbalance and overlap in industrial fault diagnosis through dynamic hierarchical sampling, integrated with a boosting ensemble to effectively combine multiple classifiers.
- (2) A novel MDP-based adaptive undersampling strategy is proposed in the outer layer to address class imbalance by dynamically selecting informative majority class samples, ensuring both diversity and representativeness.
- (3) A membership entropy-based overlap identification method and a corresponding weighted oversampling strategy are introduced in the inner layer to handle class overlap by enhancing class separability, while also mitigating residual imbalance across the fault dataset.

The rest of this paper is organized as follows. Section 2 provides a formal description of the problem addressed in this paper and gives a brief introduction to MDP. Section 3 introduces the proposed DEAHS and related details. In Section 4, the effectiveness of the proposed fault diagnosis framework is validated on the Tennessee Eastman process and a real polyester esterification process. Finally, Section 5 summarizes the main conclusions.

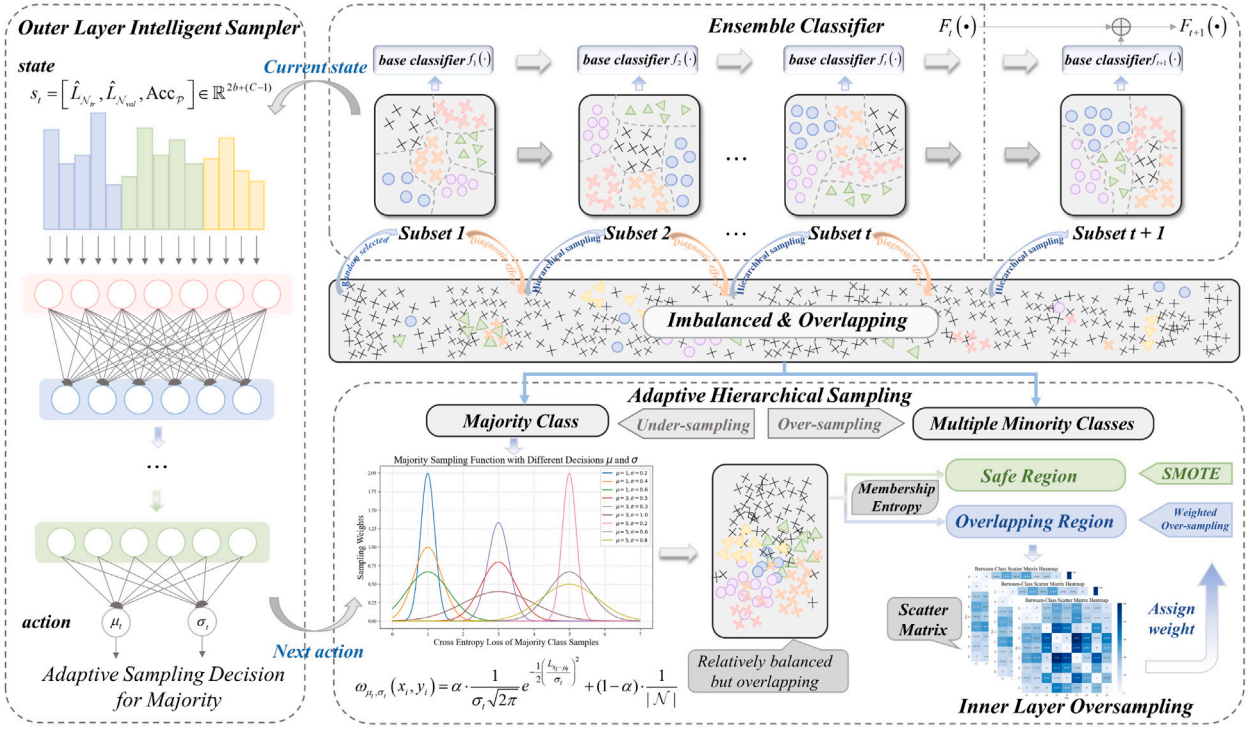


Fig. 1. Overview of the proposed DEAHS.

2. Preliminaries

2.1. Problem description

Given a multi-class imbalanced dataset $\mathcal{D}: \{\mathbf{X}, \mathbf{y}\} = \{\mathbf{x}_i, y_i\}_{i=1}^N$, where $\mathbf{x}_i \in \mathbb{R}^d$. $N = \sum_{k=0}^{C-1} n_k$ is the total number of samples in $\mathcal{D}$, n_k denotes the number of samples in class k , and C is the total number of classes. The dataset consists of a majority class $\mathcal{N}: \{(\mathbf{x}_i, y_i) | y_i = 0\}$ and multiple minority classes $\mathcal{P}: \{(\mathbf{x}_i, y_i) | y_i = 1, 2, \dots, C-1\}$, where $|\mathcal{N}| \gg |\mathcal{P}|$, and IR is defined as $|\mathcal{N}|/|\mathcal{P}|$. Furthermore, the fault data in $\mathcal{P}$ is primarily collected from early stages of faults, where their characteristics often overlap significantly with those of normal operation data $\mathcal{N}$. This overlap is quantified using the overlap ratio (OR). $\mathcal{D}$ is divided into training $\mathcal{D}_{tr}$, validation $\mathcal{D}_{val}$ and test $\mathcal{D}_{test}$ subsets while maintaining the same IR across all subsets. $\mathcal{D}_{tr}$ is used to learn a fault classifier $F: \mathbf{x}_i \in \mathbb{R}^d \rightarrow y_i \in \{0, 1, \dots, C-1\}$, while $\mathcal{D}_{val}$ is employed for hyperparameter tuning and $\mathcal{D}_{test}$ for evaluating the model's generalization on unseen data. The objective is to develop a robust fault classifier that addresses the challenges posed by the coupling of class imbalance and overlap, accurately identifying fault states, especially early-stage faults, which are critical for maintaining industrial process reliability.

2.2. Markov decision process

The MDP is constructed based on the interaction between the agent and the environment. It offers a mathematical framework for modeling decision-making in scenarios where outcomes are influenced both by chance and by the actions of a decision maker. It is typically described using a quintuple, where:

- State Space $\mathcal{S}$: The set of all possible states in the given MDP.
- Action Space $\mathcal{A}$: The set of possible actions.
- State Transition Probability Matrix $P: \{P(\bullet | s, a)\}_{(s,a) \in \mathcal{S} \times \mathcal{A}}$ is the set of one-step transition probabilities for moving to the next state s' after taking an action a in a given state s . Due to the Markov

property of MDPs, these probabilities depend only on the current state and action.

- Reward Function R : Used to calculate the reward generated by taking an action a in a given state s .
- Discount Factor γ : Used to calculate the long-term cumulative reward.

The solution to a given MDP is policy π . It is a mapping from the state space to the action space, representing the probability distribution of actions to take in each state:

$$\pi(a | s) = p(A_t = a | S_t = s). \quad (1)$$

3. Proposed fault diagnosis framework

3.1. Overview

The overall architecture of the proposed DEAHS is illustrated in Fig. 1. The framework trains an ensemble classifier using the boosting approach. In each iteration, a “balanced and well-separated” training subset is generated through adaptive hierarchical hybrid sampling.

The outer layer models the undersampling of majority class samples as an MDP to preliminarily address the imbalance in the fault dataset. The objective is to train an intelligent sampler that autonomously selects majority class samples in the next iteration based on the current classification performance of the ensemble classifier. After the outer layer sampling, the dataset's imbalance is reduced, but relative imbalance and class overlap remain. The inner layer focuses on oversampling across multiple minority classes. First, overlapping regions and safe regions are identified based on membership entropy. These regions are then processed separately, with a focus on performing weighted oversampling of minority class samples in the overlapping regions using between-class scatter matrix. The inner layer oversampling addresses class overlap while also further mitigating the imbalance. The combined effect of outer and inner layer sampling produces training subsets that are more balanced and have reduced class overlap, making them

easier for the classifier to recognize. Additionally, the framework's adaptability also enables dynamic adjustment of majority class samples, ensuring the diversity of the subsets. We employ a straightforward averaging strategy for the boosting ensemble:

$$F_t(\mathbf{x}) = \frac{1}{t} \sum_{i=1}^t f_i(\mathbf{x}) \quad (2)$$

where $F_t(\mathbf{x})$ is the ensemble classifier composed of t base classifiers, and $f_i(\mathbf{x})$ represents the i th base classifier.

3.2. Outer layer adaptive undersampling based on MDP

The outer layer of DEAHs constructs an adaptive undersampling strategy for the majority class samples to solve the imbalance in industrial fault diagnosis. It is widely recognized that ensemble learning is a common and effective strategy for addressing class imbalance issues. We model the sampling process as an MDP to create “good but different” subsets for base classifier diversity. The ensemble training process is considered as an environment, where the state is set according to the classification error of the current classifier. The sampler is trained to output the parameters of the Gaussian function, which leads to the design of an adaptively weighted undersampling strategy.

Given a sample $(\mathbf{x}_i, y_i)$, its cross-entropy loss L_{x_i} on the ensemble classifier $F_t(\cdot)$ at time step t is:

$$L_{x_i} = - \sum_{k=1}^C y_{i,k} \log(F_t(\mathbf{x}_i)_k) \quad (3)$$

The cross-entropy loss of a sample effectively measures the difference between the predicted results provided by the classification model at the current time step t and the distribution of the true labels. Therefore, we treat the sampler as an agent, using cross-entropy loss as state information to guide the next iteration of sampling. The ensemble classifier is trained via boosting, progressively enhancing the model's performance. Specifically, this process is MDP-based, with a focus on the tuple's state S , action A , and reward R , defined as follows:

- (a) State S : The concept of the error distribution histogram is introduced to illustrate the classification performance of the current model. We define the state as a combination of the cross-entropy loss histogram distribution on the majority class samples and the classification accuracy on the samples of minority classes, providing a comprehensive representation of the current ensemble classifier's performance. Formally, the cross-entropy loss on the majority sample set $\mathcal{N}$ is denoted as $L_{\mathcal{N}}$. The approximate error distribution from the histogram is represented by a vector $\hat{L}_{\mathcal{N}} \in \mathbb{R}^b$, with b being the number of bins. The j th component of $\hat{L}_{\mathcal{N}}$ is given by:

$$\hat{L}_{\mathcal{N}}^j = \frac{\left| \left\{ (\mathbf{x}_i, y_i) \mid \frac{j-1}{b} \leq L_{x_i} \leq \frac{j}{b}, (\mathbf{x}_i, y_i) \in \mathcal{N} \right\} \right|}{|\mathcal{N}|}, 1 \leq j \leq b \quad (4)$$

In addition, to capture the model's performance on minority classes, the classification accuracy for each minority class is included in the state. These accuracies are represented as a vector $\text{Acc}_p \in \mathbb{R}^{C-1}$, where $C-1$ is the number of minority classes. By combining the majority class loss histograms and the accuracies of minority classes, the state vector at time step t is defined as:

$$s_t = [\hat{L}_{\mathcal{N}_t}, \hat{L}_{\mathcal{N}_{val}}, \text{Acc}_p] \in \mathbb{R}^{2b+(C-1)} \quad (5)$$

This comprehensive state representation enables the sampler to adaptively address both majority class imbalance and minority class classification challenges, thereby improving the overall ensemble performance.

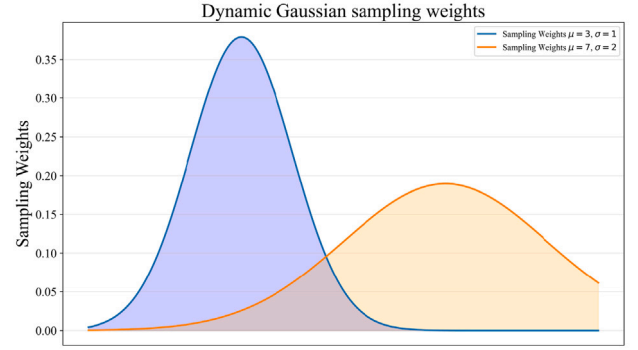


Fig. 2. Illustration of the outer layer dynamic sampling weights.

- (b) Action A : A Gaussian function is used to assign sampling weights to each sample. Therefore, based on the current state, the action output by the training sampler is the parameters of the Gaussian function, specifically the mean μ_t and standard deviation σ_t :

$$a_t = [\mu_t, \sigma_t] \quad (6)$$

- (c) Reward R : The reward at time step t is defined as the weighted change in classification metrics on the validation set after taking the corresponding action based on the current state. Formally, it is given by:

$$r_t = \sum_{i=1}^M \lambda_i \cdot [M_i(F_{t+1, \text{val}}) - M_i(F_{t, \text{val}})] \quad (7)$$

where λ_i is the weight for the i th metric, satisfying $\sum_{i=1}^M \lambda_i = 1$, and $M_i(\cdot)$ represents the i th metric used for evaluating the classification.

The sampler can be modeled as a mapping from the state space to the action space, i.e., $\pi : S \rightarrow A$. Once the sampler outputs an action, sampling weights can be assigned to the corresponding samples based on a mixture of Gaussian and uniform distributions:

$$\omega_{\mu_t, \sigma_t}(\mathbf{x}_i, y_i) = \alpha \cdot \frac{1}{\sigma_t \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{L_{x_i} - \mu_t}{\sigma_t} \right)^2} + (1 - \alpha) \cdot \frac{1}{|\mathcal{N}|}, (\mathbf{x}_i, y_i) \in \mathcal{N} \quad (8)$$

where $\alpha = 0.95$ is the mixture weight that controls the balance between focused sampling (Gaussian component) and exploration (uniform component). $|\mathcal{N}|$ denotes the number of majority class samples.

Fig. 2 shows how the outer adaptive sampler controls the sampling weights by outputting different action parameters. The action μ_t output by the sampler determine the central position of the weight distribution. When μ_t is smaller, the sampler tends to select majority class samples with smaller classification errors in the previous stage for the next base classifier. Conversely, when μ_t is larger, the sampler favors majority class samples with larger classification errors in the previous stage for the next base classifier. The output action σ_t controls the width of the weight distribution. A smaller σ_t makes the weight distribution sharper, so only majority class samples with errors very close to μ_t receive high sampling weights. In contrast, a larger σ_t results in a broader weight distribution, resulting in a more diverse selection of majority class samples. The sampler aims to find the optimal strategy through iterations of $s_t \rightarrow a_t \rightarrow s_{t+1}$, which naturally leads us to use a reinforcement learning approach based on SAC [38] to train the adaptive undersampler and solve for the optimal strategy.

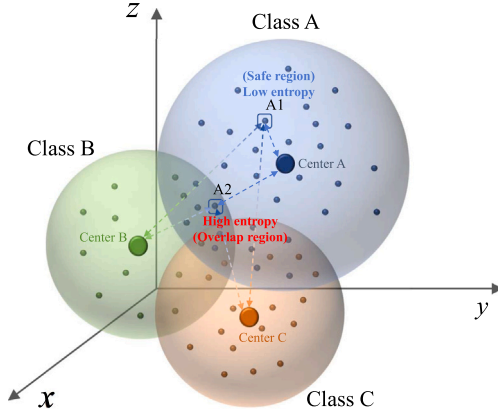


Fig. 3. Illustration of region division.

3.3. Inner layer overlapping region identification and weighted oversampling

3.3.1. Region division based on membership entropy

In complex industrial process fault datasets, different classes often exhibit overlapping samples near the decision boundaries. This overlap increases the difficulty for classifiers to accurately distinguish between classes, as these samples share similar features and have uncertain labels. In contrast, safe regions are areas where class boundaries are well-defined, making it easier for classifiers to recognize and differentiate samples. Therefore, the key to region division lies in quantifying the difficulty of classifying each sample, which allows for more targeted handling of different regions.

Therefore, the concept of membership degree is introduced. It establishes the membership relationship between samples and classes, representing the probability that a sample belongs to a particular class. The range of it is $[0, 1]$, and the sum of the membership degree of a sample across all classes equals 1. When a sample is close to a specific class and relatively distant from other classes, the membership degree is high. When the sample is close to the target class as well as to other classes, the membership degree is low. As shown in Fig. 3, point A1 is close to center A and far from center B and center C, so it has a higher membership to class A and a lower membership to class B and class C. Point A2 is similar in distance to the centers of the three classes, so its membership to the three classes is also close. The membership degree m_i^j of a sample x_i to class j is defined as:

$$m_i^j = \frac{1}{\sum_{k=1}^C \left(\frac{d_i^j}{d_i^k} \right)^{\frac{2}{n-1}}} \quad (9)$$

where d_i^j represents the distance between x_i and the center vector of class j , and n is the fuzzification parameter, typically set to 2.

Further, membership entropy $H(x_i)$ is a measure used to evaluate the uncertainty of the membership degree of x_i across all classes, and it is defined as follows:

$$H(x_i) = - \sum_{k=1}^C m_i^k \log m_i^k \quad (10)$$

Higher entropy indicates greater uncertainty in the classification of x_i , suggesting it is more likely near a class boundary or in an overlapping region. In Fig. 3, point A1 has a relatively certain high membership class, so it has a low membership entropy, corresponding to the safe region. Point A2 has a relatively vague membership, so it has a high membership entropy, corresponding to the overlapping region.

To divide the data region D , a threshold T_{ov} for membership entropy is set. The specific division strategy is as follows:

$$\{x_i \in D_{ov} \mid x_i \in D, H(x_i) > T_{ov}\} \quad (11)$$

where $T_{ov} = \mu_H + \sigma_H$ is the threshold of membership entropy. When the membership entropy of a sample exceeds T_{ov} , the sample is considered to be in the overlapping region D_{ov} . Samples in this region have unclear class assignments and may lie on class boundaries, making classification challenging. These require careful handling and further learning. The remaining samples are divided into the safe region D_{safe} , where they are easier for the classifier to identify.

$$D_{safe} = D - D_{ov} \quad (12)$$

This distinction allows for tailored handling of samples in the overlapping region, which are more challenging to classify, while safe region samples can be addressed using conventional methods. This scientifically grounded division strategy provides finer guidance for classifier training, enhancing the ability to handle overlapping samples and improving overall classification performance and robustness.

3.3.2. Weighted oversampling based on between-class scatter

Considering the highly imbalanced nature of industrial process fault datasets, excessive undersampling of the majority class can result in substantial data loss. Therefore, after dividing the overlapping and safe regions, different oversampling strategies are applied to the minority class samples within each region. This reduces dataset imbalance and makes the boundaries between classes more distinct. For the safe region, the minority class samples in the dataset are expanded using the classic oversampling method SMOTE. For the overlapping region, weighted oversampling is performed after determining the sampling weights of minority class samples based on their contribution to the between-class scatter.

First, the between-class scatter matrix S_B of the dataset is calculated as:

$$S_B = \sum_{k=1}^C n_k (\mu_k - \mu) (\mu_k - \mu)^T \quad (13)$$

where μ_k is the mean vector of class k , μ is the overall mean vector of all samples. Then, the contribution of minority class samples in D_{ov} to the between-class scatter is calculated. The contribution of x_i belonging to class j to the between-class scatter α_i is defined as:

$$\alpha_i = (x_i - \mu_j)^T S_B (x_i - \mu_j) \quad (14)$$

It represents the projection of the distance from x_i to the class center μ_j onto the between-class scatter matrix, reflecting the importance of the sample in distinguishing between different classes. The larger the value, the greater its contribution to the between-class scatter, and the corresponding oversampling weight for the sample is:

$$w_{x_i} = \frac{\alpha_i}{\sum_{x_j \in P} \alpha_j} \quad (15)$$

The training process of the proposed DEAHs fault diagnosis framework is presented in Algorithm 1.

3.4. Complexity analysis of the proposed DEAHs

The complexity of the DEAHs framework is analyzed from both spatial and temporal dimensions. Suppose the dataset contains N samples, each with d features, and there are C classes in total. The ensemble framework consists of T base classifiers, and the hidden layer dimension of the intelligent agent based on SAC is h .

In terms of spatial complexity, the storage requirements of the algorithm are divided into three main components: (1) The basic data structure requires $O(NC)$ space to store feature matrices. (2) The SAC-based intelligent agent requires $O(h(d+C))$ space to store network parameters. (3) The ensemble learning process requires $O(TM)$ space to store the T base classifiers, where M represents the parameter size of a single classifier. Thus, the total spatial complexity of the DEAHs framework is $O(Nd + TM + h(d+C))$.

Algorithm 1 The Training Process of the Proposed DEAHs

Require: Imbalanced and overlapping dataset $\mathcal{D}$, base classifier f , number of the base classifiers m , number of state histogram bins b , neural network adaptive under-sampler $AUSampler(\cdot)$

Ensure: Ensemble classifier F_m

- 1: Divide $\mathcal{D}$ into a training set $\mathcal{D}_{tr}$, a validation set $\mathcal{D}_{val}$, and a test set $\mathcal{D}_{te}$
- 2: Divide $\mathcal{D}_{tr}$ into majority class $\mathcal{N}$ and minority classes $\mathcal{P}$
- 3: Train the first base classifier f_1 with random balanced subset
- 4: **for** $t = 1, 2, \dots, m$ **do**
- 5: $F_t(\mathbf{x}) = \frac{1}{t} \sum_{i=1}^t f_i(\mathbf{x})$
- 6: Get the current state s_t according to Eq. (5)
- 7: Get the trained adaptive undersampler with output μ_t and σ_t
- 8: Obtain $\mathcal{N}'$ by weighted undersampling of $\mathcal{N}$ according to Eq. (8)
- 9: Get the relatively balanced subset $\mathcal{N}' \cup \mathcal{P}$
- 10: Divide it into safe $\mathcal{D}_{safe}$ and overlapping areas $\mathcal{D}_{ov}$ according to Eq. (11), (12)
- 11: Calculate the between-class scatter matrix $\mathbf{S}_B$ according to Eq. (13)
- 12: **if** $\mathbf{x}_i$ in $\mathcal{P}$ belongs to $\mathcal{D}_{ov}$ **then**
- 13: Calculate the weights of the samples and weighted oversampling according to Eq. (14), (15)
- 14: **else if** $\mathbf{x}_i$ in $\mathcal{P}$ belongs to $\mathcal{D}_{safe}$ **then**
- 15: Oversampling according to the SMOTE method
- 16: **end if**
- 17: Obtain $\mathcal{P}'$ by merging $\mathcal{P}$ with minority samples oversampled in $\mathcal{D}_{ov}$ and $\mathcal{D}_{safe}$
- 18: Get the balanced subset $\mathcal{N}' \cup \mathcal{P}'$
- 19: Train the new base classifier f_{t+1}
- 20: **end for**
- 21: Get the final ensemble classifier F_m

For temporal complexity, the main computational cost occurs during the adaptive training process, which includes state calculation, inner and outer layer sampling, and model training. First, the state calculation involves computing the cross-entropy loss and accuracy for each state, with a complexity of $O(NC)$. Second, the sampling process includes both outer-layer and inner-layer sampling. The outer-layer sampling has a complexity of $O(N)$. The inner-layer sampling involves calculating the membership degree, with a complexity of $O(NC^2)$, while the computation of the between-class scatter matrix has a complexity of $O(Nd^2)$. For the training phase of the framework, the training of T base classifiers is considered. The complexity of training each base classifier is $f(N, d)$, resulting in a total computational complexity of $O(I \cdot (NC^2 + Nd^2 + T \cdot f(N, d)))$, where I represents the number of adaptive iterations. Considering that the number of classes C in industrial fault datasets is usually much smaller than the number of samples N and the feature dimension d , the time complexity can be simplified to $O(I \cdot (Nd^2 + T \cdot f(N, d)))$.

The complexity analysis demonstrates that the DEAHs framework maintains a manageable computational cost when applied to industrial datasets. Specifically, as the sample size increases, complexity scales linearly, while model training (e.g., using decision trees or random forests) typically exhibits logarithmic-linear growth. This ensures the practicality of the algorithm for large-scale datasets. However, due to the quadratic dependence on the feature dimension d , preprocessing techniques such as dimensionality reduction may be necessary when dealing with high-dimensional data. These characteristics of computational complexity enable the DEAHs framework to balance efficiency and effectively address the coupling of class imbalance and overlap in industrial fault diagnosis.

4. Experiments and results

In this section, the proposed DEAHs and other imbalanced classification methods are applied to two industrial datasets: the TE process dataset and a real-world polyester esterification process dataset. A series of experiments were conducted to demonstrate the performance of the proposed fault diagnosis framework.

4.1. Experimental settings

In imbalanced learning, the commonly used classification metric, accuracy, often fails to adequately reflect the performance of the classifier. Therefore, we employ four metrics to evaluate the model's performance: Balanced Accuracy (BA), G-mean, $F1_{macro}$ and $F1_{micro}$. The formulas are as follows:

$$BA = \frac{1}{n} \sum_{i=1}^n \text{Recall}_i \quad (16)$$

$$G - \text{mean} = \sqrt[n]{\prod_{i=1}^n \text{Recall}_i} \quad (17)$$

$$F1_{macro} = \frac{1}{n} \sum_{i=1}^n F1_i \quad (18)$$

$$F1_{micro} = 2 \times \frac{\text{Precision}_{micro} \cdot \text{Recall}_{micro}}{\text{Precision}_{micro} + \text{Recall}_{micro}} \quad (19)$$

where $F1_i$ and Recall_i represent the F1-score and recall of the i th class, respectively.

We introduce the overlap ratio (OR) as a metric to quantify the class overlap in the dataset. Specifically, OR_k describes the degree of overlap for class k ; $OR_{(i,j)}$ describes the degree of overlap between class pair (i, j) ; $OR_{dataset}$ represents the overall overlap of the entire dataset. Their definitions are as follows:

$$\begin{aligned} OR_k &= \frac{n_k^{ov}}{n_k}, \\ OR_{(i,j)} &= \frac{1}{2} \left(\frac{n_{i,j}^{ov}}{n_i} + \frac{n_{j,i}^{ov}}{n_j} \right), \\ OR_{dataset} &= \frac{1}{C} \sum_{k=1}^C OR_k \end{aligned} \quad (20)$$

where n_k^{ov} is the number of overlapping samples in class k , $n_{i,j}^{ov}$ represents the number of samples in class i that overlap with class j , and $n_{j,i}^{ov}$ represents the number of samples in class j that overlap with class i .

To demonstrate the effectiveness of the proposed fault diagnosis framework, we selected five data-level methods as baseline models for comparison: SPE [36], DBCF [33], WGAN-GP [39], DDPM [40], and DeepSMOTE [24]. These methods encompass ensemble frameworks, deep learning generative models, and advanced SMOTE-based data balancing techniques. The reasons for selecting these methods are as follows:

- (1) SPE: By introducing classification difficulty metrics, this method adopts adaptive sampling to gradually focus on hard-to-classify samples while avoiding overfitting, balancing efficiency and robustness. It is suitable for addressing class overlap and large-scale imbalanced datasets.
- (2) DBCF: Integrating multi-channel deep cascade structures with hybrid sampling strategies, this approach combines data-level and algorithm-level improvements to enhance robustness and classification accuracy for imbalanced data, making it suitable for complex industrial fault classification tasks.
- (3) WGAN-GP: Built on the adversarial structure of deep neural network generators and discriminators, this method approximates distributions and applies gradient penalties to generate high-fidelity and diverse samples, ideal for addressing data scarcity and class overlap in industrial scenarios.

Table 1
Descriptions of the 28 faults in the TE process.

Fault no.	Description	Type of fault	Fault no.	Description	Type of fault
1	A/C feed ratio, B composition	Step	13	The variation of reaction kinetics	Slow migration
2	B composition, A/C feed ratio unchanged	Step	14	Reactor cooling water valve	Stick
3	D feed temperature	Step	15	Condenser cooling water valve	Stick
4	Reactor cooling water inlet temperature	Step	16–20	Unknown	Unknown
5	Inlet temperature of condenser cooling water	Step	21	A feed temperature	Random variables
6	A feed loss	Step	22	E feed temperature	Random variables
7	C pressure loss	Step	23	A feed pressure	Random variables
8	A,B,C feed composition	Random variables	24	D feed pressure	Random variables
9	D feed temperature	Random variables	25	E feed pressure	Random variables
10	C feed temperature	Random variables	26	A and C feed pressure	Random variables
11	Reactor cooling water inlet temperature	Random variables	27	Pressure fluctuation in the cooling water re-circulating unit of the reactor	Random variables
12	Inlet temperature of condenser cooling water	Random variables	28	Pressure fluctuation in the cooling water re-circulating unit of the condenser	Random variables

Table 2
Class overlap grouping in the TE process.

Group	Faults
1	Fault 1, 6, 8, 21, 23, 26
2	Fault 2, 8
3	Fault 1, 7, 10, 26
4	Fault 3, 9, 24
5	Fault 4, 11, 14, 27

- (4) DDPM: Leveraging a step-by-step noise addition and removal generation process, this approach generates high-quality and diverse data, effectively alleviating class boundary overlap, and is well-suited for handling data scarcity and complex fault distributions.
- (5) DeepSMOTE: Combining a deep encoder–decoder framework with the SMOTE algorithm, this method generates high-quality synthetic samples that enhance the separability of minority classes. Its penalty loss mechanism optimizes sample distributions, making it particularly effective for imbalanced high-dimensional data.

In the subsequent experiments, since the proposed framework is actually decoupled from the base classifier, we choose random forest (RF) as the default base classifier. This selection aligns with the framework's design, as its adaptive hierarchical sampling strategy relies on data distribution characteristics to construct diverse sample subsets for training, without depending on the specific structures or parameters of the classifiers. The modular approach ensures flexibility by enabling the framework to adapt to any ensemble-compatible classifier. Additionally, the number of base classifiers was set to 20 to maintain consistency across comparative experiments. This choice reflects a balanced trade-off between performance and computational efficiency, as we also conducted a sensitivity analysis of this parameter in Section 4.3.4.

4.2. Case 1: Tennessee Eastman process

4.2.1. Data description and preprocessing

The TE process is a widely used benchmark in the fields of fault diagnosis, optimization, control, and process monitoring. The simulation program [41] and relevant instructions used in this paper can be found and accessed at the following link: [Updated TE Code](#).

In chemical production processes, it is desirable to identify faults at an early stage and provide corresponding solutions. To better meet the

actual needs of chemical production and realistically simulate the coexistence of imbalanced data and class overlap in industrial process fault datasets, we did not use the standard TE process dataset, as its fault data is sampled when the fault mode is in stable operation. Instead, all the data were obtained using the TE simulation platform. Under normal operating conditions, 80 h of simulation were performed to obtain 2000 normal samples. For each fault condition, 100 samples were collected during the initial 0–4 h after the fault was introduced. Considering the temporal characteristics of industrial data, we appended linear timestamps to the collected samples and applied a sliding window approach with a window size of 5 and a stride of 2. Additionally, the samples within each window were flattened for subsequent analysis.

The optimized TE process involves 41 measured variables and 12 manipulated variables, and it allows for the simulation of 28 types of faults, as shown in Table 1. Faults were grouped based on prior knowledge of the TE process to better simulate class overlap. Based on five fault causes related to A, B, C, D, and reactor cooling water, the faults were categorized into five groups. Different fault types within the same group were treated as class overlaps. The detailed grouping is presented in Table 2.

4.2.2. Comparative experiments

Five groups of experiments on the TE dataset were repeated five times to enhance the persuasiveness of the results. Table 3 shows the average and standard deviation of the fault diagnosis results across four evaluation metrics for the five comparison methods and the proposed DEAHs. Additionally, to more intuitively demonstrate the stability of the methods, radar charts were plotted, as shown in Fig. 4. Analyzing Table 3 and Fig. 4, the proposed DEAHs fault diagnosis method demonstrates superior classification performance across the five sub-datasets of the TE dataset, as indicated by the higher average values and lower standard deviations on most metrics. This indicates that the proposed method performs well in terms of both accuracy and stability. Compared to other methods, our approach demonstrates particularly outstanding performance on three key metrics (BA, G-mean, $F1_{macro}$) that are crucial for imbalanced classification. This highlights the superior ability of DEAHs to account for the minority class, which aligns closely with the core principles of fault diagnosis.

The higher accuracy of the proposed method is related to the designed adaptive hierarchical sampling strategy, especially the targeted processing of samples in the overlapping area in the inner layer. The higher stability is related to the ensemble strategy introduced in the framework, which combines the results of each base classifier. The base

Table 3
Results of comparative experiments on the TE dataset.

Group	Metric	SPE	DBCF	WGAN-GP	DDPM	DeepSMOTE	DEAHS
1	BA (%)	66.21 $\pm$ 2.25	66.59 $\pm$ 4.05	36.27 $\pm$ 1.58	53.19 $\pm$ 3.26	46.61 $\pm$ 5.82	73.95 $\pm$ 3.96
	G-mean (%)	80.59 $\pm$ 1.43	80.81 $\pm$ 2.55	57.98 $\pm$ 1.23	71.61 $\pm$ 2.89	65.67 $\pm$ 4.78	85.30 $\pm$ 2.38
	F1 _{macro} (%)	66.00 $\pm$ 1.94	67.66 $\pm$ 3.55	38.64 $\pm$ 0.85	53.81 $\pm$ 2.28	48.07 $\pm$ 4.91	70.75 $\pm$ 4.20
	F1 _{micro} (%)	87.36 $\pm$ 1.15	87.36 $\pm$ 1.61	82.72 $\pm$ 0.47	86.28 $\pm$ 0.97	80.31 $\pm$ 2.41	89.46 $\pm$ 1.53
2	BA (%)	80.62 $\pm$ 4.03	84.39 $\pm$ 2.37	68.53 $\pm$ 2.25	72.16 $\pm$ 5.41	82.99 $\pm$ 5.59	87.79 $\pm$ 0.34
	G-mean (%)	88.55 $\pm$ 2.42	90.92 $\pm$ 1.46	77.63 $\pm$ 1.44	81.43 $\pm$ 5.46	88.21 $\pm$ 1.83	92.81 $\pm$ 0.27
	F1 _{macro} (%)	73.49 $\pm$ 4.54	76.99 $\pm$ 3.76	74.58 $\pm$ 2.09	74.88 $\pm$ 3.23	67.69 $\pm$ 3.54	79.48 $\pm$ 2.39
	F1 _{micro} (%)	93.15 $\pm$ 1.55	94.16 $\pm$ 1.31	95.71 $\pm$ 0.22	95.98 $\pm$ 0.50	84.11 $\pm$ 1.51	94.61 $\pm$ 0.88
3	BA (%)	74.97 $\pm$ 4.31	76.98 $\pm$ 5.15	58.93 $\pm$ 3.07	67.28 $\pm$ 2.85	55.49 $\pm$ 5.59	82.78 $\pm$ 1.01
	G-mean (%)	85.79 $\pm$ 2.51	86.98 $\pm$ 3.04	73.03 $\pm$ 2.14	79.92 $\pm$ 2.56	70.49 $\pm$ 4.28	90.27 $\pm$ 0.57
	F1 _{macro} (%)	73.60 $\pm$ 3.44	75.24 $\pm$ 3.99	64.94 $\pm$ 3.17	70.82 $\pm$ 1.82	56.50 $\pm$ 2.47	77.27 $\pm$ 1.40
	F1 _{micro} (%)	91.93 $\pm$ 0.96	92.18 $\pm$ 1.14	91.43 $\pm$ 0.63	92.94 $\pm$ 0.75	82.10 $\pm$ 2.94	92.52 $\pm$ 0.62
4	BA (%)	64.46 $\pm$ 5.21	65.41 $\pm$ 3.75	34.50 $\pm$ 1.87	49.62 $\pm$ 4.49	55.82 $\pm$ 8.25	71.03 $\pm$ 2.61
	G-mean (%)	79.46 $\pm$ 3.29	79.89 $\pm$ 2.28	54.85 $\pm$ 1.84	67.58 $\pm$ 3.27	71.85 $\pm$ 6.07	83.35 $\pm$ 1.59
	F1 _{macro} (%)	62.90 $\pm$ 5.11	61.97 $\pm$ 2.67	35.77 $\pm$ 2.55	52.31 $\pm$ 4.79	46.24 $\pm$ 4.13	66.54 $\pm$ 2.41
	F1 _{micro} (%)	92.58 $\pm$ 1.38	90.92 $\pm$ 0.20	89.00 $\pm$ 0.33	91.35 $\pm$ 0.85	81.22 $\pm$ 1.68	91.70 $\pm$ 0.62
5	BA (%)	64.71 $\pm$ 5.19	61.73 $\pm$ 6.13	34.58 $\pm$ 4.09	50.36 $\pm$ 8.40	64.63 $\pm$ 4.49	74.00 $\pm$ 1.22
	G-mean (%)	79.52 $\pm$ 3.25	77.60 $\pm$ 3.85	54.89 $\pm$ 3.88	68.95 $\pm$ 6.77	78.48 $\pm$ 2.69	85.19 $\pm$ 0.71
	F1 _{macro} (%)	59.50 $\pm$ 5.66	57.00 $\pm$ 4.42	38.10 $\pm$ 4.76	49.70 $\pm$ 7.20	55.02 $\pm$ 3.76	68.86 $\pm$ 1.28
	F1 _{micro} (%)	89.66 $\pm$ 0.97	89.08 $\pm$ 0.94	86.89 $\pm$ 0.81	89.92 $\pm$ 1.76	79.75 $\pm$ 0.86	90.84 $\pm$ 0.81

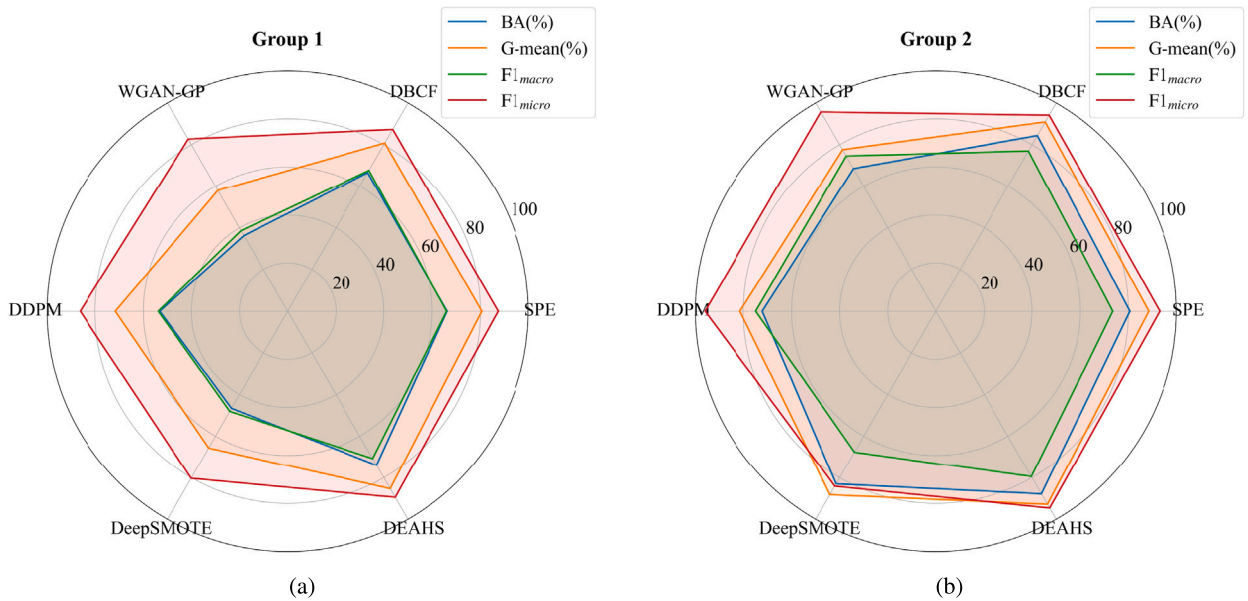


Fig. 4. Radar chart of comparative experiments on the TE dataset.

classifiers use different subsets during training, which makes the ensemble classifier have better generalization ability when facing unknown samples. Considering the imbalanced and overlapping characteristics of the experimental setup, the five comparative experiments on the TE dataset demonstrate the effectiveness of DEAHS in addressing the coupled issue of class imbalance and overlap.

4.2.3. Overlap analysis

To verify the effectiveness of the proposed DEAHS framework in addressing class overlap issues, we evaluated the inner layer sampling strategy for class overlap using OR and visualized the degree of overlap in the dataset before and after inner layer sampling. In the evaluation, random undersampling was employed to replace the outer layer sampling in each iteration, with the undersampling size set to 100. Table 4 shows the change in OR after inner layer sampling for the TE dataset Groups 1–5. Fig. 5 provides a visual analysis of the class overlap before and after inner layer sampling, using Group 1 as an example.

Quantitative analysis of class overlap shows that, despite class imbalance being a broad context, the five experimental groups on the

Table 4
Quantitative evaluation of overlap reduction using OR for Groups 1–5.

Group	1	2	3	4	5
OR (Before inner layer sampling)	21.40%	26.40%	21.10%	25.30%	28.90%
OR (After inner layer sampling)	10.40%	9.70%	7.00%	9.80%	0.80%
Relative overlap reduction	51.20%	63.40%	66.90%	61.50%	97.20%

TE dataset still face significant class overlap challenges. After inner layer sampling, the degree of overlap in the dataset was significantly reduced. In the heatmap shown in Fig. 5, darker pixel colors indicate a higher degree of overlap between classes. It is evident that after inner layer sampling, all pixel colors in the heatmap become lighter, indicating that the proposed method effectively mitigates the dataset's overlap, thereby improving the separability of the classes. The key reason for overcoming the overlap is that the proposed inner layer

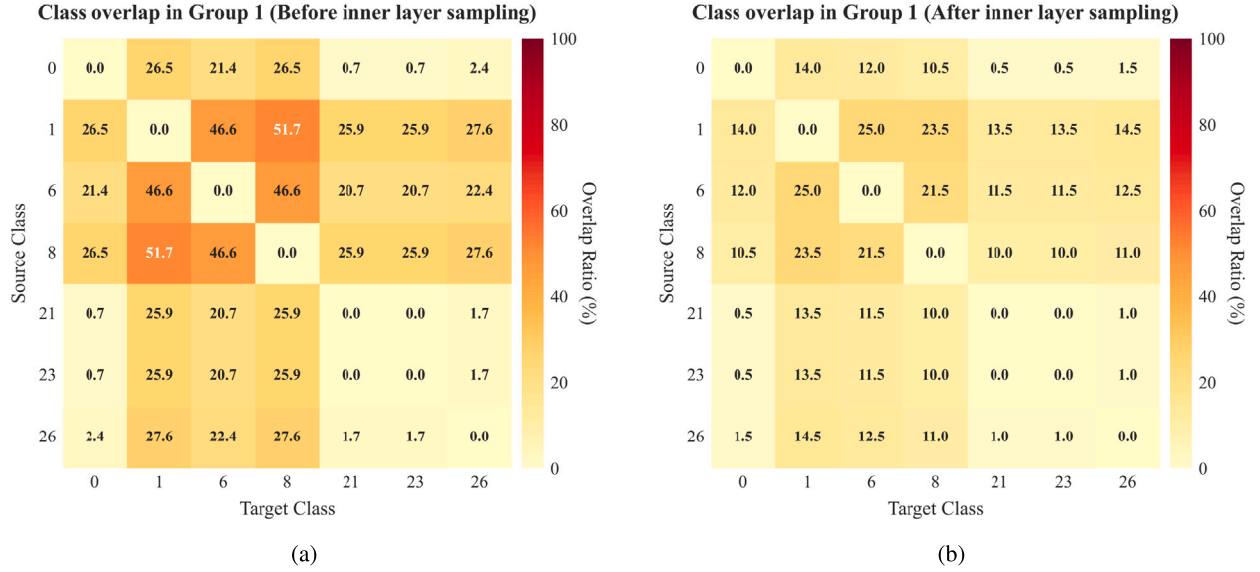


Fig. 5. Heatmap of class overlap in Group 1: before and after inner layer sampling.

sampling strategy first identifies the overlapping regions and then applies different sampling strategies to both safe and overlapping regions, significantly reducing the overlap in the dataset.

4.2.4. Noise robustness analysis

We introduced three types of noise to simulate different scenarios in real industrial systems. Gaussian noise was used to represent random measurement errors, which typically originate from sensor limitations or environmental interference in industrial systems. Salt pepper noise was applied to simulate abrupt sensor failures, such as unexpected spikes during data acquisition. Periodic noise was designed to reflect situations where noise levels vary periodically over time, representing dynamic disturbances during system operation. Each type of noise was configured with four intensity levels (5%, 10%, 15%, and 20%) to evaluate the model's robustness under varying noise conditions.

The performance of DEAHs framework under different types and levels of noise on Group 1 in the TE dataset is presented in Table 5. Despite the challenges posed by increasing noise intensity, the DEAHs framework consistently demonstrated remarkable robustness in high-noise environments. In particular, DEAHs maintained strong performance under periodic and Gaussian noise, highlighting its effectiveness in handling random errors and dynamic disturbances. Moreover, despite the greater challenges posed by high-intensity salt pepper noise, the framework effectively mitigated its impact. This robustness is attributed to the outer-layer sampling strategy, which dynamically adjusts the sampling weights of majority-class samples based on classifier performance, and the inner-layer sampling strategy, which precisely identifies and processes overlapping regions. Overall, DEAHs consistently achieved high performance across different types and intensities of noise, validating its robustness and practical applicability in complex industrial environments.

4.3. Case 2: Polyester esterification process

4.3.1. Data description and preprocessing

Esterification is a critical stage in the production of polyester fibers [42,43], involving a complex process and a variety of equipment. The flow chart of the process is illustrated in Fig. 6. The esterification reaction is carried out under high temperature and positive pressure, where various parameters such as temperature, liquid level, pressure, $n(\text{EG})/n(\text{PTA})$, and reflux can affect normal production. As this stage represents the initial phase of the polymerization process, it is crucial

Table 5

Performance of DEAHs framework under different types and levels of noise on Group 1 in the TE dataset.

Noise	Metric	Original	5%	10%	15%	20%
Gaussian	BA (%)	73.95	72.26	69.03	69.18	66.96
	G-mean (%)	85.30	84.20	82.01	82.04	80.69
	$F1_{macro}$ (%)	70.75	68.50	65.60	65.25	61.56
	$F1_{micro}$ (%)	89.46	87.21	85.27	81.98	81.40
Periodic	BA (%)	73.95	75.79	72.80	73.52	73.37
	G-mean (%)	85.30	86.41	84.54	85.00	84.87
	$F1_{macro}$ (%)	70.75	72.24	68.02	68.25	67.31
	$F1_{micro}$ (%)	89.46	89.92	87.60	88.37	87.60
Salt pepper	BA (%)	73.95	69.29	71.81	65.16	67.37
	G-mean (%)	85.30	82.52	84.00	79.80	81.36
	$F1_{macro}$ (%)	70.75	67.82	70.87	64.59	66.17
	$F1_{micro}$ (%)	89.46	88.37	89.53	87.21	88.37

to pay special attention to this stage from the perspective of industrial process fault diagnosis.

The data used in this study were collected from a real factory's DCS, where the production facility is divided into two production lines, A and B. We selected 20 monitoring variables (including 11 control variables and 9 process variables) that comprehensively represent the production conditions of the polyester esterification process, relying on production expertise. The dataset contains eight types of faults for both Line A and Line B, as shown in Table 6.

In the experiment, we selected 4,000 samples of normal operating conditions for both Line A and Line B. Each fault had 200 samples, with IR of 20:1. The data was split into a training set, validation set, and test set in a 6:2:2 ratio. To analyze the overlap between different classes in the dataset, we created ridge plots for the datasets from both production lines. The visualization showed that for the 20-dimensional features, 14 and 16 dimensions of Line A and Line B, respectively, had highly overlapping distributions. Due to space limitations, Fig. 7 displays ridge plots for four highly overlapping features from Line A: PIC_0403 A.PV, PIT_0429.PV, LIC_0434.PV, and PIT_0438.PV. The figure clearly shows that the actual production fault dataset indeed faces the dual challenges of imbalance and class overlap. Additionally, the same process step may be associated with multiple faults. For example, both Fault 2 and Fault 3 are related to the bottom EG pump, while Fault 4 and Fault 8 are associated with the esterification separation tower. Therefore, there is overlap among the faults.

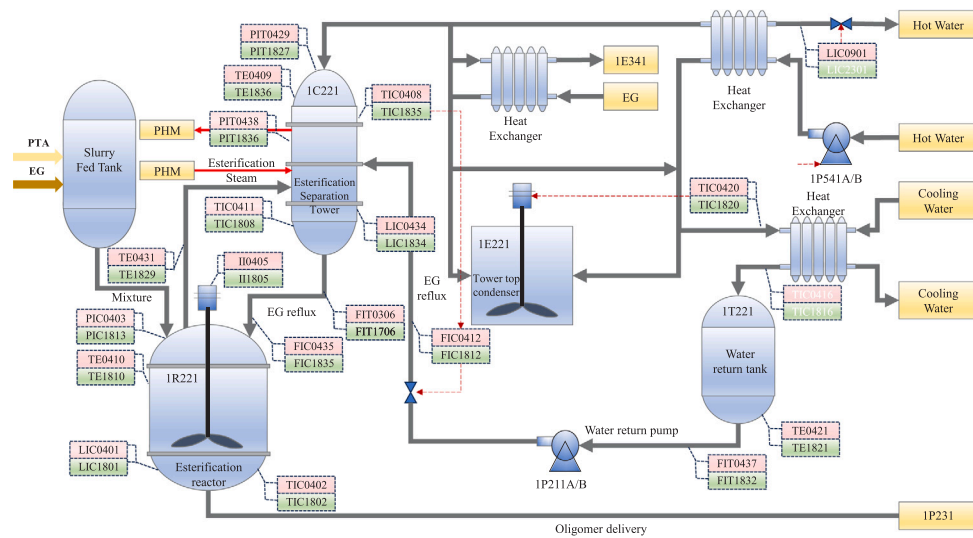


Fig. 6. Flow chart of the polyester esterification process.

Table 6
Description of faults in the polyester esterification process [44].

Fault no.	Line A-description	Fault no.	Line B-description
A1	Anomaly in the temperature control valve of the water separation in the esterification water cooler	B1	Anomaly in the temperature control valve of the separation tower
A2	Anomaly in the outlet pipeline of the bottom EG pump	B2	Anomaly in the outlet pipeline of the bottom EG pump
A3	Anomaly in the reflux pipeline of the bottom EG pump	B3	Anomaly in the reflux pipeline of the bottom EG pump
A4	Anomaly in the flow control valve of the esterification water reflux	B4	Anomaly in the flow control valve of the esterification water reflux
A5	Anomaly in the temperature control valve of the circulating hot water at the outlet of the heat exchanger	B5	Anomaly pressure in the middle of the esterification separation tower
A6	Anomaly in the level control valve of the precondensed EG	B6	Anomaly in the level control valve of the precondensed EG
A7	Anomaly in the outlet pipeline of the slurry discharge pump	B7	Anomaly in the outlet pipeline of the slurry discharge pump
A8	Abnormal pressure in the top of the esterification separation tower	B8	Abnormal pressure in the top of the esterification separation tower

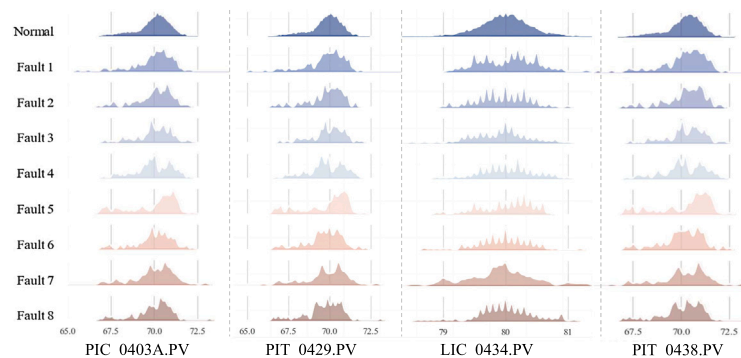


Fig. 7. Ridge plot of the four selected features in Line A.

4.3.2. Comparative experiments

The proposed DEAHs fault diagnosis framework was applied to two production lines in the real polyester esterification process, with the confusion matrix heatmap of the classification results for Line A shown in Fig. 8. Overall, the proposed method achieved better diagnostic results on Line A. Fig. 8 demonstrates that Fault 2 and Fault 3, as well as Fault 4 and Fault 8, which are highly overlapping and associated with the same process step, can be effectively distinguished.

Similar to the previous experiments on the TE dataset, we compared the proposed DEAHs with five other baseline methods and evaluated them using four metrics. All methods were repeated five times on the datasets from the two production lines, with the average and standard deviation of the comparison experiments presented in [Table 7](#). Additionally, to clearly demonstrate the robustness of the various methods, we plotted line charts with error bars for the four evaluation metrics, as shown in [Fig. 9](#). [Table 7](#) and [Fig. 9](#) show that the proposed method achieved the best performance across all evaluation metrics on

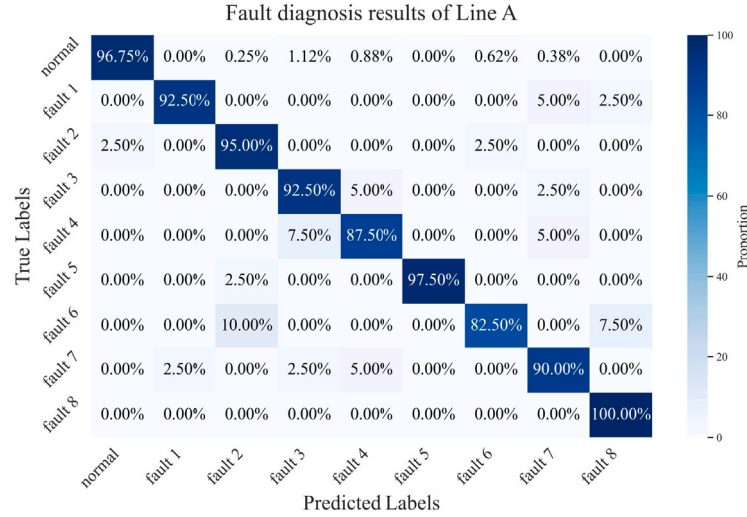


Fig. 8. Heatmap of the confusion matrix for Line A.

Table 7

Results of comparative experiments on the polyester esterification process dataset.

Group	Metric	SPE	DBCF	WGAN-GP	DDPM	DeepSMOTE	DEAHS
Line A	BA (%)	79.22 ± 1.66	78.86 ± 1.76	72.98 ± 1.59	76.58 ± 2.13	82.12 ± 1.97	94.24 ± 0.76
	G-mean (%)	88.28 ± 0.95	88.26 ± 1.00	84.33 ± 0.99	86.84 ± 1.30	89.79 ± 1.15	96.86 ± 0.40
	F1 _{macro} (%)	71.40 ± 1.43	74.21 ± 1.44	71.66 ± 1.69	77.78 ± 2.74	75.39 ± 1.44	91.62 ± 0.79
	F1 _{micro} (%)	86.22 ± 0.85	89.64 ± 0.49	85.41 ± 1.07	92.29 ± 0.69	86.84 ± 0.82	96.20 ± 0.37
Line B	BA (%)	62.44 ± 1.73	61.89 ± 1.23	64.48 ± 1.21	63.56 ± 2.84	65.44 ± 2.25	66.83 ± 0.67
	G-mean (%)	78.47 ± 1.11	78.11 ± 0.80	79.73 ± 0.77	79.17 ± 1.80	80.37 ± 1.42	81.25 ± 0.41
	F1 _{macro} (%)	61.09 ± 1.83	59.42 ± 1.72	63.48 ± 1.36	62.24 ± 2.84	64.85 ± 1.93	66.43 ± 0.67
	F1 _{micro} (%)	87.93 ± 0.56	87.75 ± 0.40	87.77 ± 0.67	88.29 ± 0.91	88.89 ± 0.72	89.34 ± 0.21

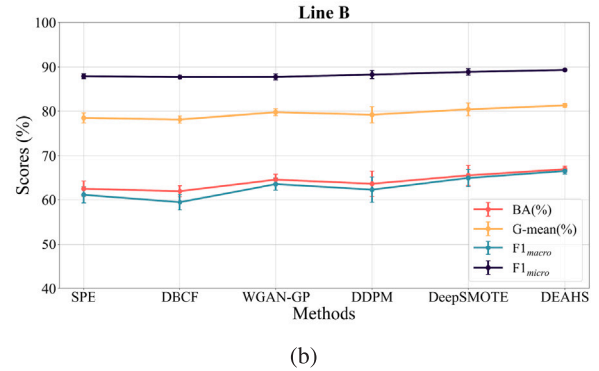
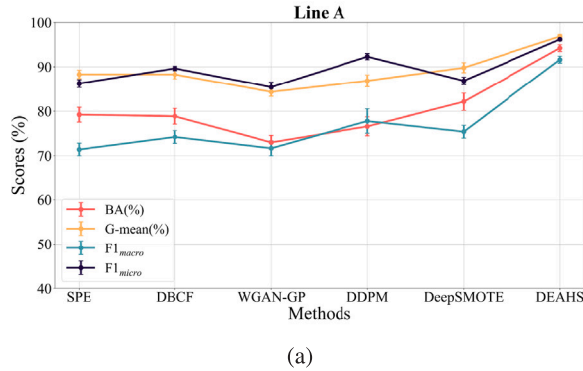


Fig. 9. Line chart with error bars of the comparison experimental results on the polyester esterification dataset.

both the Line A and Line B datasets. Notably, on Line B, where the class overlap is more severe, the proposed DEAHS still outperformed other methods on every metric. This demonstrates the effectiveness of our method in addressing the class overlap problem. The standard deviation of the experimental results also demonstrates the stability of the proposed method.

4.3.3. Ablation study

In this subsection, we conduct ablation experiments on Line A and compare the results with the complete fault diagnosis framework DEAHS to validate the effectiveness of hierarchical adaptive sampling in addressing the challenges of imbalance and class overlap coupling. Three ablation sets were configured:

- (1) Ablation 1: Both the inner and outer layer sampling were removed, with only random undersampling of the majority class performed.

- (2) Ablation 2: No identification or oversampling of the overlap region was conducted during each sampling iteration, meaning only MDP-based adaptive undersampling of the majority class was applied in the outer layer.
- (3) Ablation 3: The outer layer adaptive sampling was removed, and random undersampling was substituted before performing inner layer oversampling.

The evaluation metrics BA, G-mean, F1_{macro} and F1_{micro} were used. Each experiment was repeated five times, and the results are presented in Table 8.

In Ablation 1, both the inner and outer adaptive sampling were removed, resulting in poor fault identification performance. Specifically, the performance of the first three evaluation metrics, which focus on imbalanced datasets, is particularly poor. Combined with the analysis of F1_{macro} and F1_{micro} it is evident that the fault classifier in

Table 8
Results of ablation experiments.

Configuration	Description	Removed part	BA (%)	G-mean (%)	$F1_{macro}$ (%)	$F1_{micro}$ (%)
Ablation 1	Random undersampling	Outer and inner layer	73.75 ± 0.44	70.83 ± 0.45	70.86 ± 0.53	86.66 ± 0.32
Ablation 2	Outer layer adaptive sampling	Inner Layer	82.71 ± 2.63	79.73 ± 4.12	79.16 ± 3.34	91.00 ± 1.69
Ablation 3	Random undersampling & Inner layer	Outer layer	91.05 ± 1.14	95.12 ± 0.63	87.95 ± 1.43	94.68 ± 0.78
Proposed model	DEAHS	None	94.24 ± 0.76	96.86 ± 0.40	91.62 ± 0.79	96.20 ± 0.37

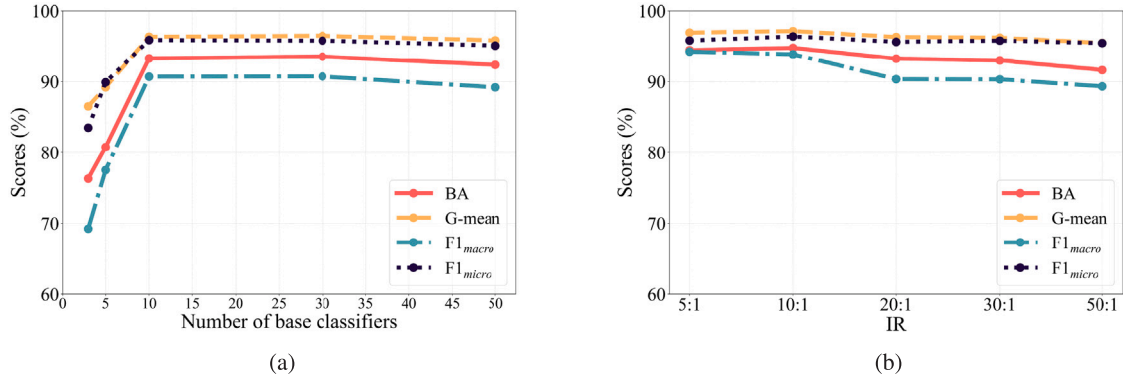


Fig. 10. Performance curves of DEAHS with varying numbers of base classifiers and IR.

this case focused more on normal data, leading to missed or false detection of fault samples, which are more critical in the production process. In Ablation 2, the first three metrics related to imbalance showed improvement compared to Ablation 1. This indicates that the outer layer MDP-based undersampling of the majority class can reduce the negative impact caused by the imbalance between normal and fault samples. However, the overall diagnostic performance still has room for improvement due to the similarities between different fault datasets. Without intervention in the minority class samples, the diagnostic model will struggle to accurately identify each type of fault. In Ablation 3, all four metrics showed improvement, indicating that the inner layer's targeted recognition and weighted oversampling of overlapping regions can effectively reduce class overlap and enhance the separability between different classes. In DEAHS, we addressed both class imbalance and class overlap, resulting in a significant improvement in fault diagnosis performance. Furthermore, the results of the ablation experiments further confirm that class imbalance is neither the sole nor the primary reason for the decline in classifier performance. The coupling of class overlap with class imbalance poses a greater challenge.

4.3.4. Sensitivity analysis

To test the robustness of the proposed method concerning the number of base classifiers in the ensemble framework and the IR of the dataset, we separately varied the number of base classifiers and IR, conducting experiments on Line A. The results are shown in Fig. 10. Fig. 10(a) shows the changes in fault diagnosis performance across evaluation metrics when the number of base classifiers in the framework is set to 3, 5, 10, 30, 50. The results indicate that the increase in the number of base classifiers exhibits a marginal effect. When the number of base classifiers increases from single digits to 10, classification performance improves significantly. There is a slight improvement from 10 to 30, but further increasing the number of base classifiers leads to a slight decrease in performance, possibly due to overfitting on the training set. In practical applications, considering the operational efficiency of the fault diagnosis framework, choosing 10 to 20 as the number of base classifiers to achieve a balance between accuracy and efficiency.

Fig. 10(b) shows the changes in the fault diagnosis performance across evaluation metrics when IR is set to 5:1, 10:1, 20:1, 30:1, and

50:1. The results indicate that the three metrics, which better measure the performance on imbalanced classifications, slightly decrease as the IR increases, while the metric representing the overall model performance, $F1_{micro}$, remains mostly unchanged. This demonstrates that the proposed model can effectively handle datasets with various degrees of imbalance. It also confirms that the increase in IR is not the sole factor leading to a decline in classification performance, factors such as class overlap play a more significant role in this process. DEAHS fully considers this factor by performing initial undersampling at the outer layer and then specifically addressing class overlap in the inner layer, thereby enhancing class separability within the dataset.

5. Conclusions

This paper proposes a dynamic ensemble fault diagnosis framework with adaptive hierarchical sampling strategy to address the coupling of class imbalance and overlap in industrial process fault diagnosis. The framework constructs different subsets through adaptive hierarchical sampling to train the ensemble model using a boosting approach. During model construction, each iteration's adaptive sampling is divided into outer-layer undersampling for the majority class and inner-layer oversampling for multiple minority classes. The outer layer tackles the imbalance issue by modeling the undersampling process as an MDP, allowing the sampler to autonomously decide which majority class samples to select for the next subset during its interaction with the classifier. The inner layer's oversampling for minority classes first divides overlapping and safe regions based on membership entropy. It then focuses on weighted oversampling of minority samples in the overlapping regions, guided by between-class scatter. This effectively reduces class overlap and further mitigates imbalance. The combined effect of inner and outer adaptive sampling effectively controls the impact of the coupling between class imbalance and class overlap on the decline in fault diagnosis accuracy. The effectiveness of the proposed framework has been validated on the TE process dataset and a real-world polyester esterification process dataset. Compared to other benchmark methods, it more accurately identifies industrial faults when faced with the coupling of class imbalance and class overlap.

Although the DEAHS framework has demonstrated strong performance in addressing the coupling of imbalance and class overlap in industrial fault diagnosis, it still has some limitations. The current

framework conducts sampling and overlap region identification primarily in the original feature space, without fully enhancing feature representation, which may constrain its effectiveness in complex industrial scenarios. Additionally, the framework lacks an incremental learning mechanism, limiting its adaptability to changes in data distribution. Future work will focus on improving feature representation, leveraging transfer learning to address data drift, and developing an incremental model update mechanism to enhance the online real-time processing of dynamic data, thereby expanding its potential for application in complex industrial environments.

CRedit authorship contribution statement

Haoyan Dong: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Chuang Peng:** Investigation, Data curation. **Lei Chen:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Data curation. **Kuangrong Hao:** Supervision, Project administration, Funding acquisition, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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